Mini-Project Guidelines & Rubric

BSE4610 Building Informatics | 20% of final course grade | Total: 100 marks

1. Submission requirements

Submission files: Executable/actionable code and a PowerPoint report.

Report length: Up to 10 content slides; cover slide(s) are excluded from this limit.

Cover slide: Clearly state your contribution to the work. If the project is completed collaboratively, identify each contributor's work.

Study location: Select a city other than Shenzhen&Tokyo for the main analysis.

Submission DDL: 26/10/26 (Week 9)

Declaration of AI Use.

2. Missions and scores

	Assessment task and expected output	Marks
1.	City selection: choose a non-Shenzhen and non-Tokyo city.	**5**
2.	Solar radiation data visualization: choose ONE of five visualization types.	**5**
3.	Plane-of-array (POA) analysis for fixed and tracking PV: select TWO of four visualization types (Section 3). Scoring: 2 types × 3 marks × 2 configurations.	**12**
4.	Comparison of daily average POA for fixed versus tracking PV.	**3**
5.	PV output calculation and visualization: compare daily average power output for fixed and tracking PV.	**5**
6.	Building analysis: building footprints with rooftop PV potential.	**5**
7.	Calculate total annual PV energy for the whole study region and plot annual energy distribution by building ID (X-axis: building ID).	**5**
8.	Calculate average energy output per square metre (m^2).	**5**
9.	Convert a selected building to a tracking PV configuration and display its annual energy distribution.	**5**
10.	Visual quality and original insights: complete ONE extension (cross-city comparison, clustering, OR prediction). See Section 4 for the 50-mark rubric.	**50**
TOTAL		**100**

Scoring note Items 1–9 = 50 marks; item 10 = 50 marks. Together they form the 100-mark mini-project score.

3. Task choices and output guidance

Item 2 | Solar radiation visualization (choose ONE, 5 marks) Solar radiation on one day; average daily solar radiation components throughout the year; monthly average daily solar radiation profiles; monthly average solar radiation components; OR seasonal average solar radiation components.

Item 3 | POA analysis (choose TWO types, 12 marks) POA components on one day; average daily POA components throughout the year; monthly average POA components; OR seasonal average POA components. Show both fixed and tracking cases for each selected type.

Presentation guidance. Use clearly labelled axes, appropriate units, informative captions, and consistent assumptions when comparing fixed and tracking systems. State data sources and main modelling assumptions so that your results can be interpreted.

4. Extension: visual quality and original insights (50 marks)

Choose ONE of the following directions. The extension should build on the core solar/PV analysis and lead to interpretable findings, not only additional figures.

A. Compare different cities. Compare solar resources or PV potential across at least two cities. Keep metrics, time periods, assumptions and units consistent; explain the main differences and likely drivers.

B. Clustering. Group buildings, time periods or other relevant observations using selected features. Explain the feature choice, visualise the resulting groups and interpret what distinguishes them.

C. Prediction. Predict a relevant solar or PV variable using an appropriate model. Explain inputs and target, evaluate performance on data not used for fitting, and discuss where predictions are reliable or uncertain.

Detailed 50-mark rubric

Criterion	What is assessed	Marks
Visual clarity and aesthetics	Clean and consistent figures; readable axes and legends; appropriate plot choices; professional slide layout and logical visual flow.	10
Question and analytical design	A clear extension objective, a meaningful comparison/grouping/prediction setup, and well-justified data, variables and approach.	10
Technical implementation	Correct and coherent analytical workflow; appropriate calculations or model implementation; code that can reproduce the extension results.	10
Depth and evidence of insights	Identify important patterns, quantify differences or performance where relevant, explain their significance, and support conclusions with results rather than description alone.	15
Critical reflection	Recognise important data or modelling limitations and outline plausible improvements or next steps.	5
	Extension subtotal	**50**

Important: Good-looking charts alone are not sufficient for full extension credit; the analysis must also provide credible, evidence-based insights.